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# Detecting and correcting false transients in calcium imaging

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**Population recordings of calcium activity are a major source of insight into neural function. Large datasets require automated processing, but this can introduce errors that are difficult to detect. Here we show that popular time course-estimation algorithms often contain substantial misattribution errors affecting 10–20% of transients. Misattribution, in which fluorescence is ascribed to the wrong cell, arises when overlapping cells and processes are imperfectly defined or not identified. To diagnose misattribution, we develop metrics and visualization tools for evaluating large datasets. To correct time courses, we introduce a robust estimator that explicitly accounts for contaminating signals. In one hippocampal dataset, removing contamination reduced the number of place cells by 15%, and 19% of place fields shifted by over 10 cm. Our methods are compatible with other cell-finding techniques, empowering users to diagnose and correct a potentially widespread problem that could alter scientific conclusions.**

Advances in understanding nervous system function have relied on large-scale optical imaging of activity at cellular resolution. As improvements in calcium imaging technology have allowed for larger, longer and more numerous recording sessions [1–8], it has become difficult to manually identify every active cell in every movie frame, making automated cell finding an indispensable tool. Automation, however, separates users from raw data, and thus it has the potential to mask errors.

The principal goal of calcium imaging analysis is to assign a fluorescence time course to each source of activity[6] (for example, cell bodies, dendrites, axons, and neuropil), and scientific conclusions often hinge on the accuracy of these time courses. Typically, the spatial extent or 'profile' of each source is first identified, either manually or automatically[3,9–12] (for naming conventions, see glossary in Supplementary Table 1). Inferring time courses from raw fluorescence, however, is automated, for example, by averaging pixels within regions of interest[9,10] or using more complex algorithms[3,11].

When source profiles overlap, they can cause mutual time course contamination, and therefore most methods aim to tease apart activity in overlapping sources[13–16]. In this work, we focus on contamination from non-neuropil sources, such as overlapping dendrites or cell bodies. Although evaluation metrics have been developed to check for contamination[10,11,17], they tend to be global rather than individual, and thus are unable to assess individual transients. Here we investigate the origin and effects of time course-estimation errors in two-photon imaging and develop tools to quantify and remedy such errors. We first examine a brain region where errors are particularly likely to occur: the CA1 region of the hippocampus, where cell bodies are densely packed with thick apical dendrites[13]. We closely examine source data to motivate quantitative metrics to identify contamination and then introduce a time course-estimation algorithm to explicitly model and filter out contaminating signals. We show that our method eliminates over 90% of contaminating activity while preserving over 90% of true activity.

## Results

**False transients occur and can impact scientific conclusions.** We first considered how accurately time courses capture fluorescence in an example dataset from the dorsal CA1 (ref. [18]). We identified source profiles using constrained non-negative matrix factorization (CNMF)[19] and estimated time courses by fitting source profiles to each frame independently with least squares[19].

In one patch containing 52 identified sources (Fig. 1a), the profiles resembled cells in the average image, which suggests that the algorithm accurately identified real cells. Similarly, the source time courses matched the expected statistics of calcium dynamics in CA1: a baseline punctuated by transient increases (Fig. 1b). To evaluate whether these transients correctly correspond to activity, we examined the original movie in the spatial and temporal vicinity of each transient. Specifically, we computed the 'transient profile' (a weighted average of the transient frames) to show which movie pixels gave rise to the transient. Transient profiles often resembled the source profile (Fig. 1b, transients 1–4), confirming correct attribution ('true transients'), but some transient profiles bore no resemblance to the source profile (Fig. 1b, transients 5–8), indicating misattribution ('false transients').

For a more complete assessment, a human expert examined all 33,090 transients across 1,325 sources using a custom graphical interface (Extended Data Figs. 1 and 2 and Supplementary Figs. 1–3). Most transients were confidently classified as true (67%) or false (13%). Others (20%) fell into more complex categories to be considered later. Many identified sources (66%) contained at least one false transient, and, for some sources (9%), the majority of transients were false (Fig. 1c). We also observed contamination in inferred spikes (Fig. 1c). The amplitude distributions of true and

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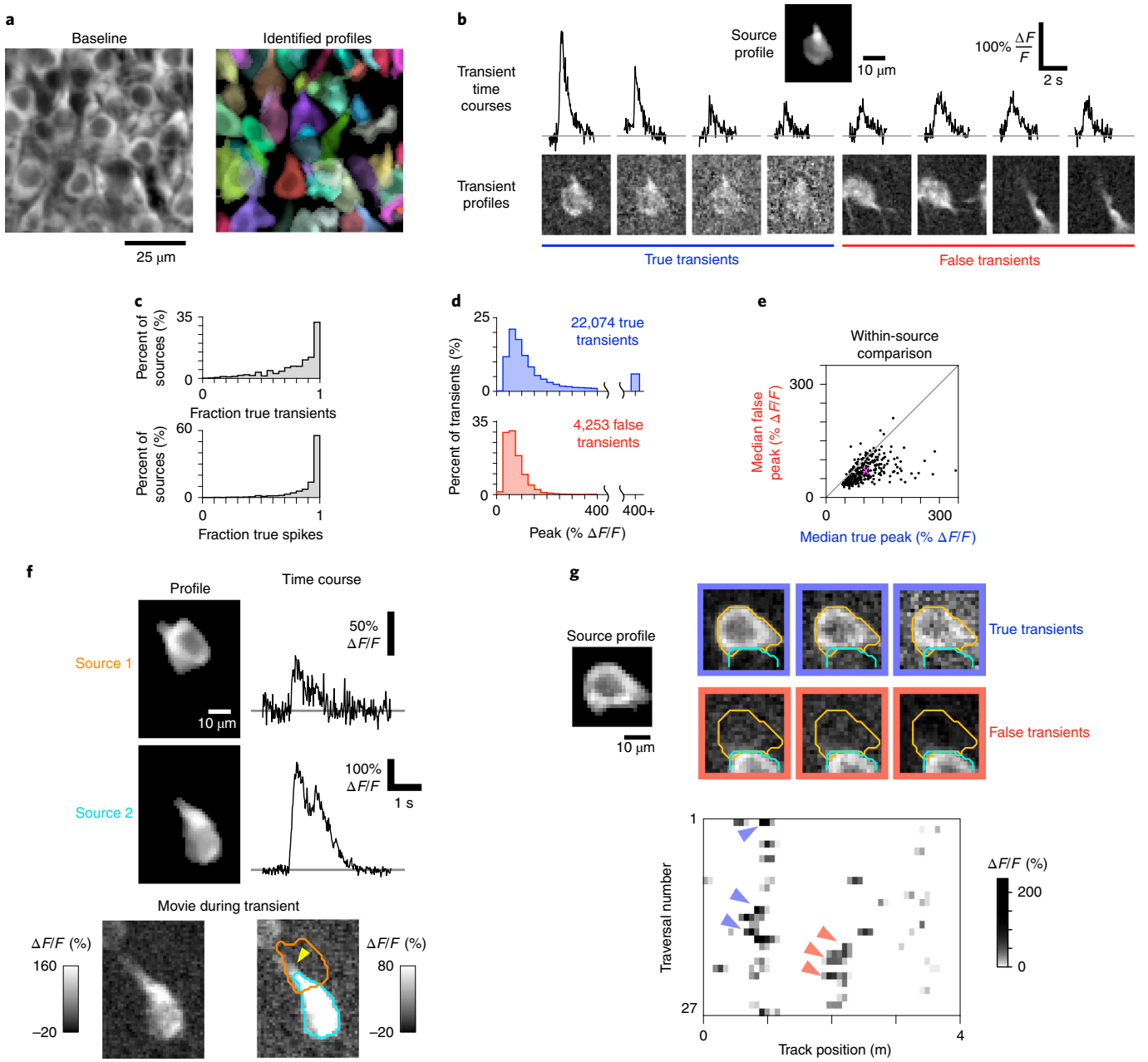



**Fig. 1 | False transients occur and can interfere with scientific conclusions. a**, Left, mean fluorescence image from a subset of an in vivo two-photon imaging movie showing mouse CA1 pyramidal cells genetically expressing the calcium indicator GCaMP3. Right, spatial profiles of the 52 sources identified by CNMF in this movie, each in a different color. **b**, Time courses (top) of eight transients assigned to one source (top inset image) and their respective transient profiles (bottom). Classification as true or false was performed manually by a human expert. **c**, Occurrence of true activity in a set of 1,325 simultaneously recorded sources. Top, the fraction of transients that were classified as true (only for sources with at least one classified transient). Bottom, equivalent data for spikes that were inferred from classified transients as returned by CNMF (only for sources with at least 1,000 spikes inferred from classified transients). **d**, Peak amplitudes of true and false transients for the same sources as in **c**. **e**, Median peak amplitude of true and false transients for each source, only shown for sources with at least five true and five false transients. The purple 'X' indicates the mean over sources. **f**, Top and middle, two source profiles and their respective time courses during a brief period of the movie. Bottom, transient profile for source 2 (left) and the same image shown with increased contrast and overlaid outlines of source profiles (right). The arrowhead indicates part of source 2 fluorescence that was not captured by its profile. **g**, Top, six transient profiles (right) assigned to the same source (left). Outlines show profiles from this source and an adjacent source. Bottom, spatially averaged activity of this source while the mouse traversed a virtual linear track. Arrowheads indicate the transients shown above.

false transients were largely overlapping across all sources (Fig. 1d) and within each source (Fig. 1e). Manual classification was comparable between different graders (Extended Data Fig. 3).

To understand the extent of misattribution, we examined transients found by several algorithms: CNMF with different parameter settings (Supplementary Fig. 10), Suite2P[3] (Supplementary Fig. 11), in-house source localization from the Allen Institute run on data from the Mouse Brain Observatory (Supplementary Table 3 and https://help.brain-map.org/display/observatory/Documentation) and semi-automated source localization[20] (Supplementary Table 3).

Contamination levels varied with labeling density, source shapes and optical properties, with greater contamination generally occurring when labeled cells were densely packed and near major crossing dendrites. Despite this trend, contamination was consistently 15–20% across all methods and datasets that we tested, with the exception of the sparsely labeled cortex. Thus contamination is likely to be widely prevalent, including in published studies of large-scale calcium imaging.

False transients can affect scientific conclusions in several ways. First, cross-talk can produce spurious correlations in activity, creating a misleading picture of network organization (Fig. 1f). Second, contamination can change apparent coding properties. For example, place cells are typically active in a small part of the environment location[21], such as a virtual linear track[18]. An important question in place coding is how many discrete fields each cell maintains[22], but false transients can create additional firing fields (Fig. 1g). To help mitigate the effects of false transients, we turn next to developing tools to identify and remove contamination.

**Quantifying contamination.** We first developed summary measures to approximate the level of contamination within a dataset. Distinguishing true and false transients is based on the assumption that the pattern of bright pixels in a movie frame should look similar to the source profile during a true transient but not during a false transient. We measured similarity by taking the Pearson correlation between the source profile and the transient profile (Fig. 2a,b). As expected, true transients have a higher correlation than false transients, and a single threshold generally provides accurate separation, even across different sources within the same movie (Fig. 2c).

Although this measure can distinguish transients that are clearly true or false, more complex activity patterns also occur. For example, true and false activity can occur simultaneously in 'mixed' transients (Fig. 2d). In addition, some source profiles identified by cell finding are artifacts that do not correspond to any source in the movie. These can arise from motion or the merging of multiple sources (Fig. 2d, bottom). Our measures aim to classify both kinds of errors as false transients, although it should be noted that some true activity is consequently rejected. The correlations of mixed and artifact transients are generally similar to those of false transients (Fig. 2e and Supplementary Fig 4); therefore, the correlation threshold is reasonably accurate for these categories. Additional metrics, for example, two-dimensional (2D) Ljung–Box quantile (LBQ) tests of residual spatial independence, can trade off better artifact rejection for lower overall performance (Extended Data Fig. 4).

To overcome these limitations, we also developed a secondary quantification. If the 'contamination rate' is the fraction of transients classified as false, we define the 'contamination severity' as the ratio of unexplained fluorescence to total fluorescence. Contamination severity allows us to distinguish the various causes of false classification (Fig. 2a,b,e). Plotting contamination rate versus severity visually summarizes misattribution across sources (Fig. 2f and Supplementary Fig. 5). Each plot region supports a particular conclusion about the quality of source estimation. The lower–left quadrant contains sources that are largely accurate. The lower–right quadrant indicates a subtle mismatch in estimating the source profile, implying low contamination. Sources located in the upper sections suffer more severe contamination, and the horizontal axis indicates whether contamination is rare (left) or frequent (right). This visualization, the frequency and level of contamination (FaLCon) plot, thus provides a bird's eye view of contamination in a dataset or across datasets (Methods).

**Geometric intuition for contamination filtering.** While the metrics developed thus far can give an overall sense of contamination and the software tools allow detailed manual inspection and classification, these alone are incomplete solutions. Both require parsing activity into discrete transients, which is not always appropriate or possible, and manual inspection is not entirely reproducible and often prohibitively time consuming. Moreover, any keep-or-reject scheme cannot untangle the true and false activity within mixed transients. To address these shortcomings, we introduce an automated time course estimator that models contamination explicitly to remove it on a frame-by-frame basis. Before fully describing this model, we briefly explore a geometric intuition for how false transients can be removed.

Consider a two-pixel movie (Fig. 3a). Each frame is a point in 2D, and each source profile shape is a line through the origin. A true transient would be a point near the line (Fig. 3b), while a false transient would be far away (Fig. 3c). Assuming independent, identically distributed Gaussian noise, the optimal activity estimator finds the profile scaling (that is, point on the line) that minimizes the summed squared error (Fig. 3d). For a true transient, least squares produce a good fit. For a false transient, however, the added fluorescence erroneously pushes the fit away from zero.

This error can be avoided by reducing the cost of positive residuals[23], creating a skewed least-squares cost function (Fig. 3e). This skewed cost function reproduces the least squares fit to a true transient, while not registering the false transient. Having illustrated the fundamental principle underlying our approach, we now construct an explicit model of contamination and develop an estimator to prevent it from affecting source time courses.

**Sparse emulation of unknown dictionary objects model and algorithm.** Beginning with the standard model, we add a term for fluorescence that does not arise from known cell shapes (Fig. 4a). This unexplained fluorescence is modeled as a sum of narrow Gaussian kernels with independent, exponentially distributed amplitudes. Linear superpositions of these kernels can capture the shape of almost any contaminating fluorescence. This procedure, which we name sparse emulation of unknown dictionary objects (SEUDO), removes contamination by absorbing unexplained fluorescence into 'SEUDO cells' built of Gaussian kernels.

Mathematically, we denote the observed fluorescence image $y(t)$ in frame $t$ restricted to the spatial location of neuron $n$ as

$$y_n(t) = \Phi a(t) + b(t) + \epsilon(t), \tag{1}$$

where $a(t) \geq 0$ is the vector of activation levels for known sources, $\Phi$ denotes their spatial profiles, $b(t) \geq 0$ is the image of structured contamination and $\epsilon(t)$ is the unstructured noise vector. We consider each frame $t$ and neuron $n$ independently, and so omit these indices for clarity. The contamination $b$ is assumed to be non-negative, spatially correlated and spatially sparse and to be non-zero with probability $P$. Thus we model $b = Wc$, where $c \geq 0$ is sparse (that is, mostly zeros) and $W$ represents a dictionary of positive 2D Gaussian kernels that induce spatial correlations.

Our model motivates the estimator

$$\widehat{a} = \arg\min_{a} \min\{f(a), g(a)\}, \tag{2}$$

where $f(a)$ is the least-squares estimator for a and $g(a)$ minimizes the partial least absolute shrinkage and selection operator (LASSO)

$$g(a) = \min_{c \geq 0} \left[ ||y - \Phi a - Wc||_2^2 + \alpha ||c||_1 + \beta \right], \tag{3}$$

for constants $\alpha = 2\sigma^2\lambda$ and $\beta = -2\sigma^2 \left( \log(\frac{1}{P} - 1) + N \log(1/\lambda) \right)$ and where $N$ is the number of pixels, $\lambda$ is the mean parameter for the exponential distribution of $c$ and $\sigma^2$ is the noise variance. The estimate minimizes $f(a)$ and $g(a)$ separately and then takes the value of a at the smaller of the two. SEUDO thus resembles a likelihood test on the presence of unexplained fluorescence. Visualizing

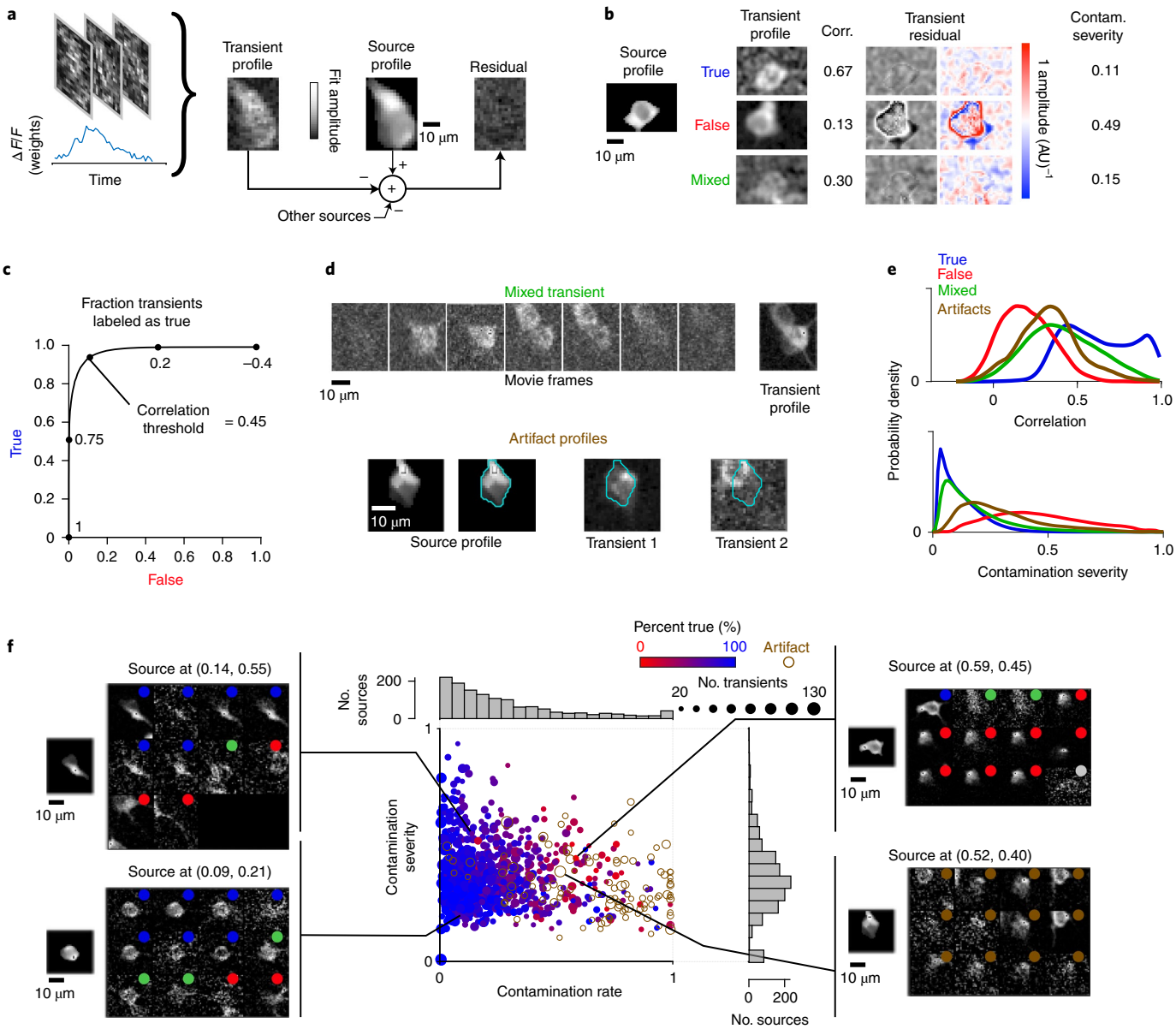



**Fig. 2 | Quantitative metrics for evaluating transients. a**, Schematic of how the transient profile and transient profile residual are computed. **b**, Three transients (rows) for one source; the source profile is at the far left. Shown for each transient are the human classification (true, false or mixed), the transient profile with Gaussian blurring to highlight structure (radius 1 pixel), the correlation (corr.) metric value, the transient residual (shown with two color maps) and the contamination (contam.)-severity metric. AU, arbitrary units. **c**, Performance of automatic classification using the correlation metric. **d**, Top, example of a transient with mixed activity (movie frames are shown as three-frame averages). Bottom, example of an artifact profile arising from the fusion of two cell bodies. Left images show source profile with and without contour, and right images show two representative transient profiles with contour overlaid. **e**, Distribution of correlation values (top) and contamination severity (bottom) for transient categories defined by human classification. **f**, Center, FaLCon plot for 1,325 sources identified in one movie of CA1 (same dataset as shown in Fig. 1a) at a correlation threshold of 0.5. For axis definitions, see text and the Methods. Each point is one source; color shows human classification. Left and right, for each of four example sources, images show the source profile (single panel) and all transient profiles (grouped panels). Inset dot colors reveal human classification of each transient (gray indicates transients that could not be confidently classified).

the SEUDO estimator in the case of a two-pixel movie (Fig. 4b–e) shows that it matches the skewed cost function described earlier (Fig. 3e), providing an intuition for how SEUDO is able to remove contamination while preserving true activity.

As with many algorithms, SEUDO is strongly affected by parameter choice. The parameters are the Gaussian kernel radius, the probability of contamination $P$, the exponential parameter $\lambda$ and the noise variance $\sigma^2$. The kernel radius, which determines the spatial scale of contamination, can be estimated directly from the movie frames. The three remaining parameters appear as only two coefficients: $\alpha$ and $\beta$ (equation (3)). Effectively, $\alpha$ is the per-kernel cost and $\beta$ is a constant cost whenever kernels are used. If the overall cost of kernels is sufficiently high, they are not used and SEUDO reverts to least squares (Extended Data Fig. 5a, right). If the cost is sufficiently low, kernels are always used, diminishing or eliminating all transients (Extended Data Fig. 5a, left). In an optimal regime, true transients are retained, while false transients are removed (Extended Data Fig. 5a, center). Because SEUDO is relatively insensitive to

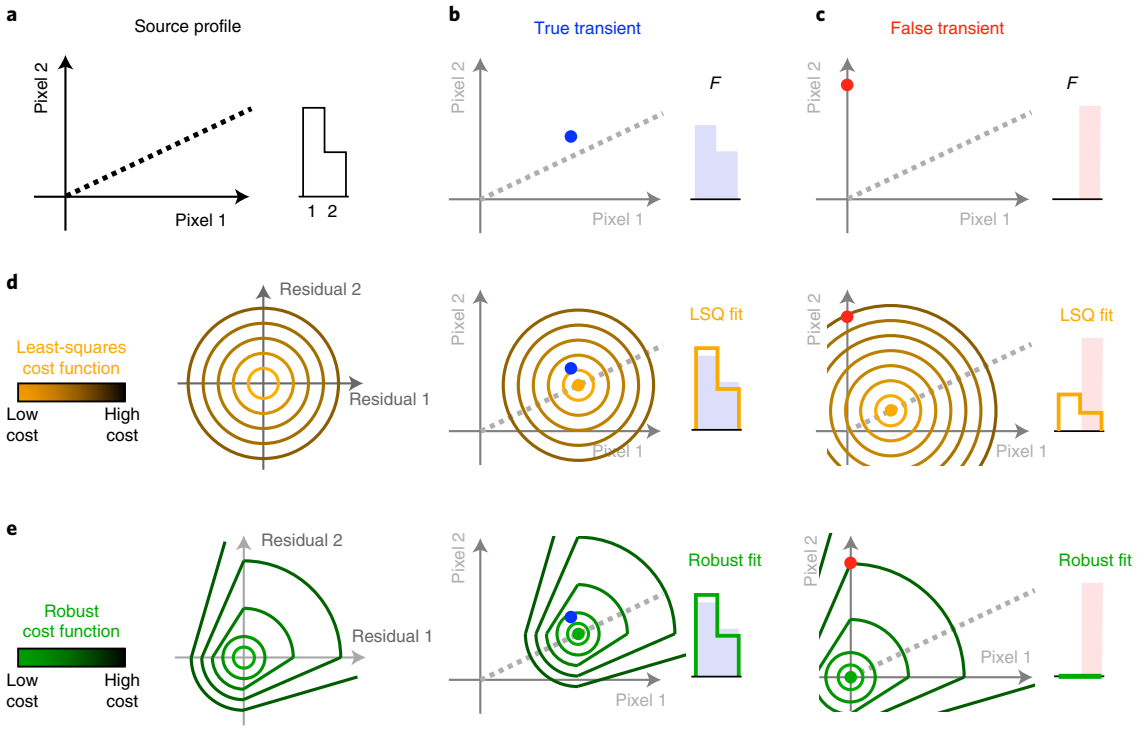



**Fig. 3 | Geometric intuition for the proposed solution. a**, Visualization of a simulated source profile in a two-pixel movie. Left, dashed line shows points corresponding to possible scalings of this profile. Right, brightness versus pixel number for one scaling of this profile. **b**, A true transient represented as a point (left) and a plot of brightness (right). **c**, Equivalent plots for a false transient. **d**, Left, contour lines of the least-squares (LSQ) cost function. Middle and right, best fit of the source profile (gold point and gold outline) using least squares. **e**, Left, contour lines of a hypothetical robust cost function. Middle and right, best fit of the source profile (green point and green outline) using the robust cost function.

$P$, we sweep $\lambda$ and $\sigma^2$ for a pair of example transients to reveal the optimal regime (Extended Data Fig. 5b). To facilitate this process in other datasets, we have developed several tools for manual or automatic parameter selection (Methods).

**Application to CA1 data.** We next applied SEUDO to CA1 data to observe its effects on neural coding properties (Fig. 1). Running SEUDO on one source (Fig. 5a) almost completely eliminated false transients, while true activity remained largely intact. Generally there is a trade-off between removing false transients and preserving true transients. We illustrate SEUDO's trade-off for this source by varying $\lambda$ while holding $\sigma^2$ constant, sweeping out a receiver operating characteristic (ROC)-like curve (Fig. 5b). Because some applications are insensitive to transient amplitude, we also considered transient detection, finding that SEUDO achieved 100% accurate discrimination for this source (Fig. 5b).

To evaluate SEUDO more generally, we applied the same parameters to the entire movie, in which we classified 26,327 transients from 1,149 sources as true or false with high confidence. For the best $\lambda$ value, >90% of false transients were rejected, while >90% of true transients were retained, demonstrating that parameters generalize well across sources. Varying $\sigma^2$ produced nearly identical discrimination, albeit with different optimal $\lambda$ values (Extended Data Fig. 6). We observed similar results for profiles and time traces from Suite2p[3] (Supplementary Fig. 9; 96.6% of false transients were removed, and 91.8% of true transients were retained).

As a baseline for comparison, we also performed decontamination using framewise Pearson correlations. For transient detection, SEUDO provided superior separation of true and false activity (Fig. 5c). When measuring preserved transient area, however, performances of the two algorithms were more similar (Supplementary Fig. 8), which can be reconciled by observing that transient area can be reduced while still preserving transient shape and duration (for example, Fig. 5a). Importantly, the correlation criterion cannot distinguish mixed transients, a limitation of any accept-or-reject classification.

An advantage of SEUDO is the ability to model mixed transients. In one example (Fig. 5d), the source was initially active (frame 20), but before its activity completely decayed, two other overlapping cell bodies became active (frames 30–50). The least-squares time course includes activity of both the source and the contamination. The SEUDO time course, however, more accurately estimates the source activity, as verified by manual inspection of source-only (cyan) and contamination-only (magenta) pixels.

Finally, we revisited the effect of contamination on scientific conclusions (Fig. 1f,g). For the case of spuriously high correlation ($r = 0.333 \pm 0.09$, 95% confidence interval; $P < 1 \times 10^{-36}$, two-sided test for non-zero correlation, comparing all 41,756 frames), SEUDO removed all but a minimal trace of residual correlation ($r = 0.018 \pm 0.1$, 95% confidence interval; $P < 0.0005$, two-sided test for non-zero correlation; Fig. 6a). Removing false correlations also had global effects, qualitatively altering standard characterizations of neural populations, including the clustering alignment and the spectral decay of the data matrix (Extended Data Fig. 7).

For the place cell with the additional false spatial field (Fig. 1g), SEUDO removed the false field almost completely (Fig. 6b) while maintaining the true field. When considering all cells with significant place fields ($P < 0.05$, test for significant spatial modulation, Methods) before and after SEUDO (Fig. 6c and Supplementary Fig. 13), 17.5% were no longer place cells after decontamination and 18 place cells were not found until false transients were removed. Moreover, an additional 18.7% of place cells had their fields shift substantially (>10 cm), further changing the characterization of the neural population.

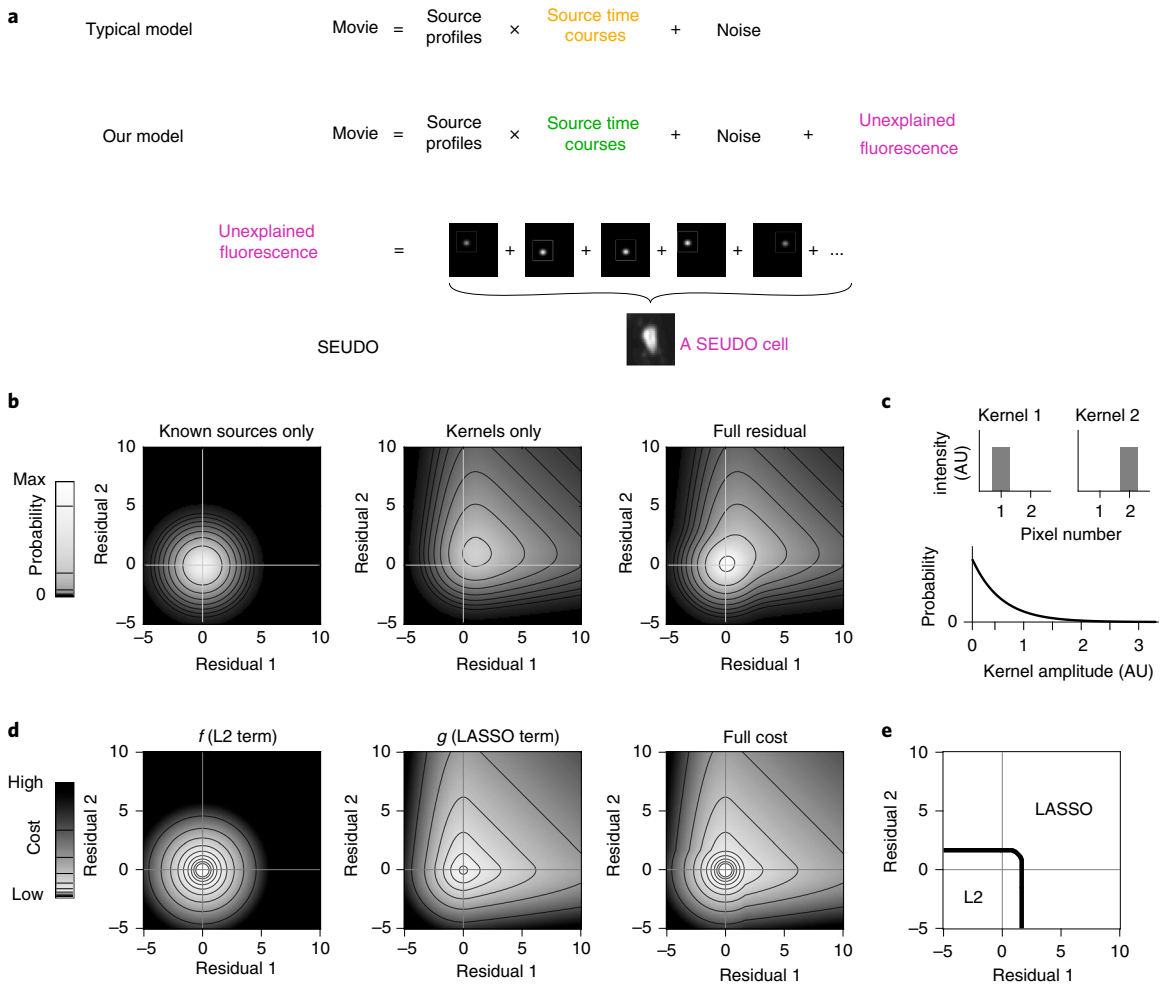



**Fig. 4 | Explicit model of unexplained fluorescence. a**, Schematic of two models for calcium imaging data. In a typical model, source time course weights (gold) are estimated using only known profile shapes. In our model, time course weights (green) are estimated using a procedure that accounts for unexplained fluorescence (magenta). **b**, Probability distribution of the SEUDO residual computed via numerical integration (unexplained fluorescence plus noise) in the case of a two-pixel movie. Left and center, individual components. Right, full residual. Max, maximum. **c**, Gaussian kernel spatial profiles (top) and amplitude distribution (bottom). **d**, Calculated cost for the objective function defined by SEUDO. Left and center, individual terms. Right, full cost; the boundary between the $\ell_2$ term and the LASSO term ($f$ and $g$ in equation (2)) is the lower cost. **e**, The full cost in **d** displays LASSO-like behavior for large positive residuals and least-squares-type behavior for small and negative residuals.

## Discussion

Automated cell finding has enabled researchers to access the rich information available in two-photon calcium imaging. Here, we introduce algorithms and software to verify and correct the time courses extracted from large movies. To obtain a global impression based on discretized transients, summary plots and coarse classification can be used to approximate the extent of contamination. In cases in which contamination is present, we provide several avenues for addressing it. When only a few cells need to be corrected or checked, a software interface allows for rapid manual classification. For correction of a large dataset or to ensure reproducibility, we further developed a robust time course-estimation technique, SEUDO. Because SEUDO operates on individual frames, it does not require discretizing activity into transients, and, by modeling contamination, it can tease apart mixtures of true and false activity.

Importantly, SEUDO is a post hoc analysis and thus can be combined with other cell-finding procedures (for example, refs. [3,10–12,24]). In cases in which time courses and source shapes are identified together[3,19], future development might allow SEUDO to be embedded directly into the algorithm.

To make SEUDO easier to adopt, we aimed to minimize the need for parameter tuning and to optimize code execution. For FaLCon plots, users only specify one parameter, the correlation threshold, which can be estimated from a small test set. Similarly, SEUDO requires only two parameters beyond typical preprocessing. We further provide parameter-selection tools via receiver operating characteristic curve visualization and BayesOpt[25]. Because SEUDO can be performed on each cell and frame independently, our software defaults to analyzing small spatiotemporal segments in parallel, resulting in, for example, a runtime of <5 min for a block of 52 cells over 41,756 frames (23 min at 30 Hz).

We have also characterized the limitations of SEUDO. Although SEUDO cannot generate false transients, it can err in two ways: reducing true activity or failing to remove false activity. The latter occurs when there is a mismatch in the contamination statistics assumed by SEUDO, as in the case of neuropil, although other algorithms have been developed to address neuropil contamination. Alternatively, SEUDO may reduce true activity when source profiles do not match observed activity. This can occur at low-light levels or when profile shapes change over time, for example, due to axial

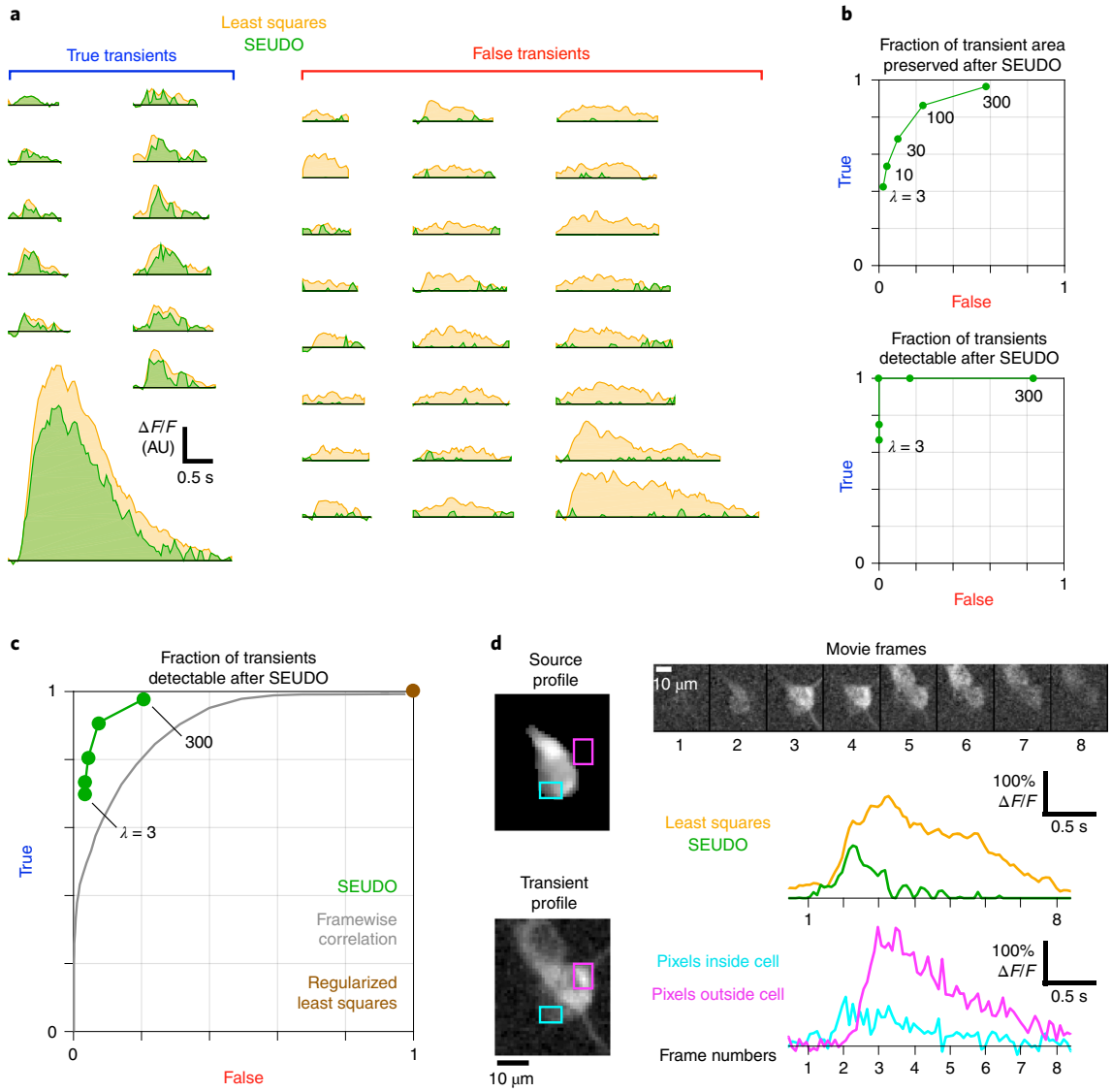



**Fig. 5 | SEUDO performance on a full population of sources. a**, Comparison of 36 transient time courses estimated by least squares (gold) or SEUDO (green, $\lambda = 30$, $\sigma^2 = 0.002$). Within each category (human-classified true or false), transients are sorted by duration. **b**, Summary of SEUDO performance for five $\lambda$ values when applied to the transients shown in **a**. **c**, Performance of SEUDO (green), a simple framewise correlation metric (gray) and regularized least-squares time course estimation (brown). SEUDO was performed using the same parameter sets as in **b**. The correlation metric was applied using 101 correlation thresholds ranging from −1 to 1 (Methods). The regularized least-squares method was CNMF. **d**, Detailed analysis of one transient with overlapping true and false activity. Shown movie frames are downsampled tenfold to highlight when different cells become active. Cyan and magenta traces are the mean pixel value in the respective rectangular regions shown at the left. The SEUDO time course was computed using the same parameter set as in **a**.

drift. Thus, for extended recordings, SEUDO should be applied separately per session to ensure source profiles are stationary.

SEUDO differs from past methods in two important ways. First, the interference model $b = Wc$ is high dimensional ($\dim(c) = \dim(b) = N$) and flexible, as the elements of $W$ are smaller than neurons and can thus adapt to any shape. The dimensionality and flexibility requires the regularizing sparsity penalty over $c$ in $g(a)$ to constrain the solution. Similar additive terms (for example, Suite2p, CNMF–CaImAn) are low dimensional ($\dim(c) \ll \dim(b)$) and lack the flexibility to adapt to false transient shapes. The second difference is the additive parameter $\beta$. While a constant in the optimization, $\beta$ does help to preserve the true transient shapes. Setting $\beta$ to 0 lowered the rates of false transient rejection (maximum decrease of ~0.06 in false transient rate) and of transient area preservation (maximum decrease of ~0.15; Supplementary Fig. 18). $\beta$ has this effect by ensuring that SEUDO cells are not used if the noisy data shape is similar enough to the profile shape, thereby preventing unnecessary explanation of variance via a nonexistant $Wc$.

These considerations point to what might seem like a paradox: because kernels can construct any shape, what is to stop them from explaining known source profiles? Compressive sensing has explored sparse estimation using over-complete dictionaries and provides theoretical accuracy guarantees on the LASSO estimator[26]. Essentially, two structures can be demixed (profiles and kernels) if they are oriented in sufficiently different directions in fluorescence space. Because kernels are highly localized, they can be distinguished from source profiles when the latter are sufficiently large.

Errors in annotating large datasets are not limited to calcium imaging. Electrophysiologists have long faced similar challenges

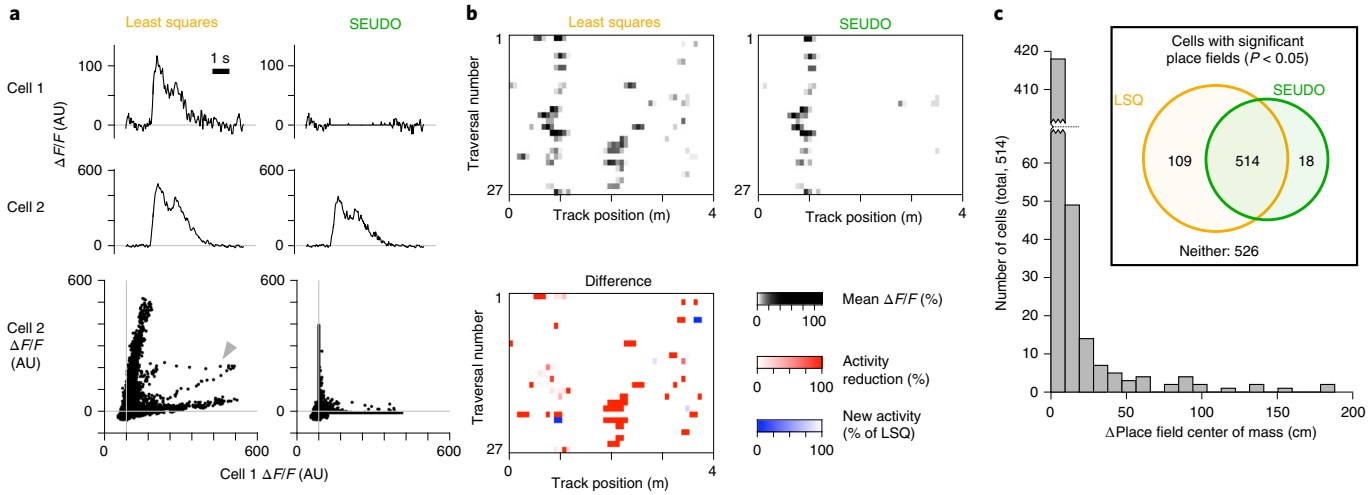



**Fig. 6 | SEUDO can correct errors affecting scientific results. a**, Top and middle, time courses for two simultaneously recorded cells with mutual contamination as estimated by least squares and SEUDO. Bottom, scatterplot comparing time courses of the same two cells over all 41,756 frames, shown separately for each estimation technique. **b**, Top, spatially averaged activity for the source depicted in Fig. 1g, shown separately when the time course was estimated by least squares or SEUDO. Bottom, the difference between least squares and SEUDO. **c**, Histogram, changes in place cell location after applying SEUDO. Inset, number of cells characterized as place cells ($P < 0.05$, test for significant spatial modulation, Methods) using time traces obtained by least squares or SEUDO.

in spike sorting[27–31], and it has become standard practice to include metrics for contamination. Two-photon imaging studies might also benefit from similar practices.

SEUDO could likely be improved or extended in several ways. For example, framewise SEUDO does not include priors over the spatial and temporal properties of contamination, nor does it account for neuropil[23,32]. To better distinguish neural activity from noise, temporal smoothing or additional spatial structure in $Wc$ could be introduced by modifying $f(x)$ and $g(x)$ in equation (2). More broadly, there might be applications in other domains for modeling contamination as non-negative structured interference (for example, one-photon calcium imaging[23], spike sorting[31] or hyperspectral imagery[33]).

## Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41592-022-01422-5.

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## Methods

**Experimental procedures and data acquisition.** All experiments were performed in compliance with the Guide for the Care and Use of Laboratory Animals (https://www.aaalac.org/resources/theguide.cfm). Specific protocols were approved by the Princeton University Institutional Animal Care and Use Committee.

To obtain large-scale population recordings of activity, transgenic mice expressing genetically encoded calcium indicators were surgically implanted with a chronic window that allowed optical access to the region under study. To image in CA1, the implant was designed as described previously[13]. At the time of surgery, mice were 7–15 weeks of age, and all mice were male. Mice were maintained on a reverse light cycle, and imaging was performed during the dark phase. In holding rooms, ambient temperatures were 69–72 °F with 30–70% humidity. GCaMP3 expression in CA1 was obtained with the mouse line C57BL/6J-Tg (*Thy1*-GCaMP3) GP2.11Dkim/J (Jackson Laboratory strain 028277)[34], and expression of the calcium indicator GCaMP6f in CA1 was obtained by crossing two mouse lines: Ai93-D;CaMKII$\alpha$-tTA (*Igs7*$^{tm93.1(tetO\text{-}GCaMP6f)Hze}$ Tg(Camk2a-tTA)1Mmay/J, Jackson Laboratory stock 024108) and *Emx1*-Cre (B6.129S2-*Emx1*$^{tm1(cre)Krj}$/J, Jackson Laboratory stock 005628). To image in the cortex, we obtained expression of GCaMP6f by crossing three strains: *Emx1*-Cre (B6.129S2-*Emx1*$^{tm1(cre)Krj}$/J, Jackson Laboratory stock 005628), CaMK2-tTA (B6.Cg-Tg(Camk2a-tTA)1Mmay/DboJ, Jackson Laboratory stock 007004) and TITL-GCaMP6f (Ai93;B6.Cg-*Igs7*$^{tm93.1(tetOGCaMP6f)Hze}$/J, Jackson Laboratory stock 024103).

Activity was measured using two-photon laser scanning microscopy employing resonant galvanometer scan mirrors at a frame rate of 30 Hz. Raw data were processed and collected using ScanImage version 5 (Invidrio). To measure spatially modulated activity in CA1 neurons, mice were trained to run on a virtual linear track[18].

**Identifying source profiles and transient times.** *CA1 dataset labeled with GCaMP3.* Motion correction and source profile identification were performed as described previously[18]. Briefly, the movie was spatially divided into a grid of 36 submovies, and each was analyzed separately. Source profiles were identified using CNMF[19] (MATLAB, 2016 version) or Suite2p[3] (MATLAB, downloaded 2 April 2019, version 1.0.0.0), and source time courses were computed using least squares (see below).

Because of small axial deflections[18], a motion artifact time course was subtracted from each source time course before filtering (median length, 10) and thresholding (zeroing points less than four times the robust standard deviation). Any contiguous period of above-threshold activity lasting at least six frames (0.2 s) was considered a transient. These same steps were applied to identify transients across all algorithms.

*Datasets labeled with GCaMP6f.* Subpixel motion correction was performed via either local (CA1 dataset) or global (MMA dataset) correlation-based shifts[11]. Source profiles and time courses were identified using CNMF[19]. For each source time course, two candidate thresholds were computed: the 95th percentile of time course values, and 11 times the robust standard deviation. Of these two values, the larger was used. Any contiguous period of above-threshold activity lasting at least 15 frames (0.5 s) was considered a transient.

**Least-squares time course estimation.** Most automated calcium segmentation methods estimate time courses via a regularized least-squares method[3,10,19,24]. If $Y \in \mathbb{R}^{N\times T}$ is the fluorescence movie with $N$ pixels and $T$ frames and $\Phi \in \mathbb{R}^{N\times M}$ is the matrix containing the spatial profiles for all $M$ isolated sources, we compute the $M \times T$ matrix of per-source fluorescences $A$ as

$$\widehat{A} = \arg\min_{A} \left[ \| Y - \Phi A \|_F^2 + R(A) \right], \tag{4}$$

where $R(A)$ is the regularization that imparts additional information about the expected time course structure. We seek to bypass the idiosyncrasies of each algorithm by focusing on time course estimation based purely on the Gaussian noise model without regularization, that is, least-squares fitting:

$$\widehat{A} = \arg\min_{A} \left[ \| Y - \Phi A \|_F^2 \right]. \tag{5}$$

**Transient profiles.** To calculate the transient profile, consider the fluorescence movie data $Y$ cropped to the spatial extent of the source profile and to the temporal extent of the transient. We then seek the best representation of $Y$ that has the same time course as the source's transient activity over the duration of the event, which we denote as $h$. This is accomplished by solving the optimization program

$$x_{\text{tp}} = \arg\min_{x} \| Y - xh^T \|_F^2, \tag{6}$$

that is, we are looking for the spatial pattern that best projects the time course into the data. The solution to this optimization program $x_{\text{tp}}$ is the transient profile. The solution to this optimization is $x_{\text{tp}} = Yh / \| h \|_2^2$, that is, the weighted sum of frames where the $i$th frame is weighted by the $i$th element of the transient time course, normalized by the total magnitude of the transient event.

**Manual classification.** Manual classification was performed by human experts (A.S.C. and J.L.G.) visually comparing the shapes of source profiles and transient profiles. Transient profiles that contained the same shape as the source profile but not other sources were classified as true. Those containing only sources different from the source profile were classified as false. Transients with a mixture of true and false transients were classified as mixed. Transients in which no spatial structure was visible or that appeared ambiguous were left unclassified. Sources for which the profile shape did not match any transient profiles (for example, fused sources) or sources that appeared to arise from a motion artifact were labeled as artifacts. To better resolve edge cases, small spatial blurring was applied at times to highlight low-amplitude structures.

For the CA1 dataset labeled with GCaMP3 as well as the dataset recorded in the visual cortical area MMA, all sources were manually classified. For the CA1 dataset labeled with GCaMP6f, 10% of sources were randomly selected for manual classification.

**Contamination severity.** The contamination-severity metric seeks to quantify the fraction of fluorescence not explained by the source profile. We thus define two residuals for every transient. The first is the full residual, which is the result of subtracting the contributions of all source profiles (columns of the pixels-by-sources matrix $\Phi$) from the pixels-by-frames fluorescence movie $Y_{ik}$ during the $i$th transient of source $k$:

$$r_{ik} = \text{vec}\left[ Y_{ik} - \Phi \widehat{A}_i \right], \tag{7}$$

where $A_i$ is the sources-by-frames matrix of estimated temporal activity for each source and each transient frame. The second is the residual subtracting off the contributions of all the sources except the source profile in question, that is, removing the column of $\Phi$ and the row of $A_i$ corresponding to the source that the transient was assigned to:

$$\widetilde{r}_{ik} = \text{vec}\left[ Y_{ik} - \widetilde{\Phi}\widetilde{A}_{ik} \right] = \text{vec}\left[ r_{ik} + \phi_k a_{ik}^T \right], \tag{8}$$

where $\widetilde{\Phi}_k = [\phi_1, \ldots, \phi_{k-1}, \phi_{k+1}, \ldots, \phi_M]$ and $\widetilde{A}_{ik} = [a_{i1}, \ldots, a_{i(k-1)}, a_{i(k+1)}, \ldots, a_{iM}]$ are the sets of all other sources and all other time courses, respectively, $\phi_k$ is the profile for the $k$th source and $a_{ik}$ is the time course for the $k$th source during the $i$th transient. The contamination severity computes the amount of residual energy in all suspect transients (that is, those that fail the correlation or the LBQ test) as a fraction of the residual discounting the source:

$$RR_k = \frac{\sum_{i\in\Gamma} \| r_{ik} \|_2^2}{\sum_{i\in\Gamma} \| \widetilde{r}_{ik} \|_2^2} = \frac{\sum_{i\in\Gamma} \| r_{ik} \|_2^2}{\sum_{i\in\Gamma} \| r_{ik} \|_2^2 + \sum_{i\in\Gamma} \| \phi_k a_i^T \|_F^2}. \tag{9}$$

When the residual is small and the source is active, the contamination severity is low; however, when the source is not active with respect to the residual size (that is, significant activity remains), then the contamination severity is high. To make the contamination severity more resistant to noise, a low-pass Gaussian filter with one-pixel standard deviation is applied.

**Spatial Ljung–Box quantile test.** The LBQ test is a statistical test of whether a model with $d$ degrees of freedom captures the autocorrelation structure of a time series, if $x(t)$ is the time series and $r(t)$ is the model residual. If $x(t)$, $r(t)$ have length $N$, and $c(l) = \text{corr}(r(t), r(t-l))$ is the autocorrelation at lag $l$, then the LBQ test calculates the quantile statistic

$$Q = N(N+2) \sum_{l=1}^{L} \frac{c(l)}{N-l}. \tag{10}$$

If $r(t)$ values are not correlated, then $Q$ should have a $\chi^2$ distribution with $L$ degrees of freedom. Consequently, a simple quantile test can be used to determine whether we can significantly rule out this null hypothesis. Specifically, we test whether $Q > q(1-\zeta, d)$ for a given significance level $\zeta$, which, if true, rules out the null hypothesis. For spatial data, that is, $x_{i,j}$, $r_{i,j}$ of size $M \times N$, we can expand the test statistic to be

$$Q = MN(MN+2) \sum_{(l,k)\in\Gamma} \frac{c_{k,l}}{(M-k)(N-l)}, \tag{11}$$

where $c_{k,l}$ is the spatial autocorrelation function at lags $k$ and $l$, and $\Gamma \subseteq [1, M] \times [1, N]$ are the set of lags to be tested. With this change, the quantile test above remains the same, aside from adjusting the degrees of freedom to account for the increased dimensionality. For all plots in this work, we set $d$ to be the number of active pixels in the profile and test the set of lags between three and seven lags inclusively. This latter choice ensures that blurring from the motion-correction process (which can blur neighboring pixels together) will not bias the results. Finally, we run all the tests at a significance level of $\zeta = 0.01$.

**Frequency and level of contamination plots across datasets.** The rate-versus-severity plot can also provide comparisons across datasets, for example, comparing to densely labeled CA1 data with GCaMP6f (Supplementary Fig. 6) and sparsely labeled cortical neurons (Supplementary Fig. 7). Compared to Fig. 2f, it is obvious that the second CA1 dataset suffers from even more extreme contamination, while the cortical dataset suffers little contamination (Supplementary Table 2).

**SEUDO algorithm motivation.** We seek to use the observed movie frame $y$ and source profiles $\Phi$ to identify the source activity $a$ (that is, the time course value in this frame). Accounting only for the known source profiles, the least-squares estimate of a is

$$a_{\text{profiles}} = \arg\min_{a} ||y - \Phi a||_2^2.$$

The SEUDO algorithm also considers an alternative case, one in which the movie frame contains fluorescence from unknown sources in addition to the known profiles. To prevent unknown sources from contaminating the estimate of $a$, SEUDO fits a set of 2D Gaussian kernels, $W$, multiplied by a vector comprising each kernel's activation level $c$. The fit also incorporates cost terms $\alpha$ and $\beta$ to impose sparseness, as unknown sources are expected to be spatially compact shapes such as cell bodies or neurites. Thus the estimate of $a$ given by fitting both profiles and kernels is as follows:

$$\{a_{\text{profiles and kernels}}, c\} = \arg\min_{a,c\geq 0} ||y - \Phi a - Wc||_2^2 + \alpha||c||_1 + \beta.$$

SEUDO then uses the estimate with the lowest cost:

$$a_{\text{SEUDO}} = \begin{cases} a_{\text{profiles}} & \text{if} \quad f(a_{\text{profiles}}) \leq g(a_{\text{profiles and kernels}}) \\ a_{\text{profiles and kernels}} & \text{if} \quad f(a_{\text{profiles}}) > g(a_{\text{profiles and kernels}}) \end{cases}$$

where $f(\cdot)$ and $g(\cdot)$ are the least-squares cost and the LASSO cost, respectively. We note that this procedure can be written as

$$a_{\text{SEUDO}} = \arg\min_{a} \min\{f(a), g(a)\}.$$

We now expand these terms to provide a more detailed optimization. In the first case, we assume that no contamination occurs; therefore, the maximum-likelihood estimator can be written as

$$a_{\text{profiles}} = \arg\max_{a} \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left(-\frac{||y - \Phi a||_2^2}{2\sigma^2}\right) \tag{12}$$

$$= \arg\min_{a} ||y - \Phi a||_2^2 + 2\sigma^2 \frac{N}{2} \log(2\pi\sigma^2). \tag{13}$$

In the latter case, we must estimate both the known profiles' activity $a$, along with the unknown and structured contamination $Wc$. To model this unknown term, we assume that $W$ is a dictionary of 2D Gaussian kernels and $c$ is a set of weights, one for each kernel. To achieve sparseness and non-negativity, we model the multivariate exponential $c$ with parameter $\lambda$. For simplicity, we will also make $W$ square, so that the dimensionality of $b$ and $c$ are the same (that is, one kernel per movie pixel). We can consider an estimator in this case as a mixed maximum a posteriori and the maximum-likelihood estimator (or a maximum a posteriori estimator with an improper uniform prior over $a$) as

$$\{a_{\text{profiles and kernels}}, c\} = \arg\max_{a,c\geq 0} \frac{\lambda^N}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left(-\frac{||y-\Phi a - Wc||_2^2}{2\sigma^2}\right) e^{-\lambda\|c\|_1} \tag{14}$$

$$= \arg\min_{a} ||y - \Phi a - Wc||_2^2 + 2\sigma^2\lambda \parallel c\|_1 + 2\sigma^2 \tfrac{N}{2} \log(2\pi\sigma^2) - 2\sigma^2 N \log(\lambda). \tag{15}$$

These two sub-estimators motivate the form of the SEUDO algorithm, where we fit each model and then compare the cost functions of each estimator to find the 'better fit'. In this way, because the term $2\sigma^2(\frac{N}{2})\log(2\pi\sigma^2)$ appears in both cost functions, it drops out of the final SEUDO cost. We further penalize the use of using the estimator that includes both a and $c$ that can be interpreted as a log probability of contamination. Combining all these terms results in the SEUDO estimator, with constants $\alpha = 2\sigma^2\lambda$ and $\beta = -2\sigma^2\left(\log(\frac{1}{P} - 1) + N\log(\lambda)\right)$. This interpretation allows the SEUDO parameters to be approximately expressed as model parameters that can be estimated more readily from data (for example, the noise variance).

SEUDO is closely related to log-likelihood ratio tests. In essence, the likelihoods of the model with and without SEUDO cells are compared (Gaussian likelihoods that result in least-squares losses), and the L1 norm over the SEUDO coefficients $c$ stand in place of, for example, the number of coefficients used as a penalizing term in information criteria such as the AIC or the BIC.

**SEUDO implementation.** As SEUDO is a convex optimization very similar to a constrained LASSO, we can implement SEUDO using well-studied first-order optimization methods[35,36]. To incorporate the specifics of this optimization, namely the non-negativity of $c$ and the fact that $a$ is unregularized, we leverage the fast and flexible Templates for First Order Conic Solvers package (version 1.4.1)[36]. To compensate for the high-dimensional model in the inference, SEUDO can be run locally for each profile independently. This reduces both the overall dimension of the vector $c$ to infer and enables parallelization over profiles.

**Performing SEUDO.** To evaluate SEUDO, we applied it to the CA1 dataset labeled with GCaMP3. SEUDO was performed separately on each of the 36 blocks. Each source was analyzed separately, and only for movie frames in which a substantial transient was identified using least squares. Pixels were only included for analysis if they fell within a rectangular region centered on the source profile; the rectangle was defined as the minimum bounding rectangle plus a radius of three pixels in all directions. Except when otherwise indicated, parameters for SEUDO were $\lambda = 10$, $\sigma^2 = 0.002$ and $P = 0.00001$, and the Gaussian kernel radius was equal to 3. Before performing the SEUDO procedure, the movie was slightly denoised by taking a rolling average of three frames. To run SEUDO on the binary profiles returned by Suite2p, we first used the binary profiles and time traces to infer the non-binary spatial profile maps for each component.

**SEUDO parameter optimization.** To facilitate parameter selection, we provide several tools, both supervised and unsupervised, to help identify the optimal parameter regime for a given dataset. First, we provide software that enables rapid manual classification of transients, building up an example set that can be used to automatically identify optimal parameters, either through an exhaustive parameter search or using BayesOpt[25].

**Estimating the Gaussian spatial kernel width.** We have implemented estimation of the Gaussian spatial kernel in two ways: a simple heuristic and via optimization. For the heuristic, we take Fourier transforms of individual movie patches and estimate the width of the resulting frequency transform's diameter. Under the sum-of-Gaussian kernel model, the frequency transform should contain a Gaussian shape with a width the inverse of the Gaussian kernel width. Thus we fit a mixture of two Gaussian shapes to the data to account for the small spatial frequency and the more uniform noise (wide in frequency). The narrow diameter represents the inverse of the Gaussian kernel diameter in space. The second method used BayesOpt[25], which jointly optimizes over all parameters. Both methods resulted in comparable estimates, and so we opted for the simple, computationally efficient heuristic.

**Framewise correlation decontamination algorithm.** As a baseline, time courses were decontaminated using the profile correlations. For each source, a small spatial subset of the movie was considered. In each frame, the Pearson correlation was computed between the movie and the source profile. If this correlation fell below threshold, the time course was set to 0. The threshold was varied from −1 to 1, and transient detection was applied to the time course.

**Transient metrics.** Two quality measures were used to evaluate the various time course estimates (least squares, SEUDO, least squares decontaminated with framewise correlation and regularized least squares). The only frames considered were ones in which a substantial transient was observed in the least-squares estimate. Results were averaged over all true transients and all false transients separately (as per expert classification).

The first measure was the fraction of transient area preserved in comparison to that of least squares. If $\frac{\left(\sum_t a^+_{\text{test}}(t)\right)}{\left(\sum_t a^+_{\text{lsq}}(t)\right)}$ is the rectified time course estimated by the technique being tested, and $a^+_{\text{lsq}}(t)$ is the rectified least-squares time course, the fraction of transient area preserved is

$$\frac{\left(\sum_t a^+_{\text{test}}(t)\right)}{\left(\sum_t a^+_{\text{lsq}}(t)\right)}.$$

The second measure simply evaluated whether the transient was still present. In this case, transient detection was applied to the entire time course. If at least one transient occurred in the period of the least-squares transient, then that transient was considered kept. Otherwise the transient was considered rejected.

**Place cell statistics.** Whether a cell exhibited a spatially modulated field was defined by how much information its activity provided about the animal's location in the environment[18]. Briefly, the observed activity was randomly shuffled many times. The amount of spatial information from each shuffle was accumulated into a null distribution, and, if the observed activity had sufficiently high spatial information (within the top 5% of shuffles), it was considered to be a place cell.

**Reporting Summary.** Further information on research design is available in the Nature Research Reporting Summary linked to this article.

## Data availability

The data that support the findings of this study, including the CA1 raw imaging data, identified source profiles and time traces from automated cell-finding algorithms and manual annotations of transients, are available on the Open Science Framework at https://osf.io/zt54q/.

## Code availability

All algorithms described are available in a MATLAB-based software suite. The software also includes several graphical user interfaces (GUIs) to facilitate visualization and parameter optimization. For example, one GUI displays transients for manual classification (Extended Data Fig. 2). Users can also interact with this GUI to specify which transients are true, false or mixed or identify artifact sources. Other GUIs (not shown) facilitate comparing different sets of parameters to optimize performance. All software is available under the MIT license and can be downloaded from https://github.com/adamshch/SEUDO.

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## Author contributions

J.L.G. and A.S.C. developed and applied analysis methods; J.L.G., S.A.K. and E.H.N. performed imaging experiments; all authors wrote the paper.

## Competing interests

The authors declare no competing interests.

## Additional information

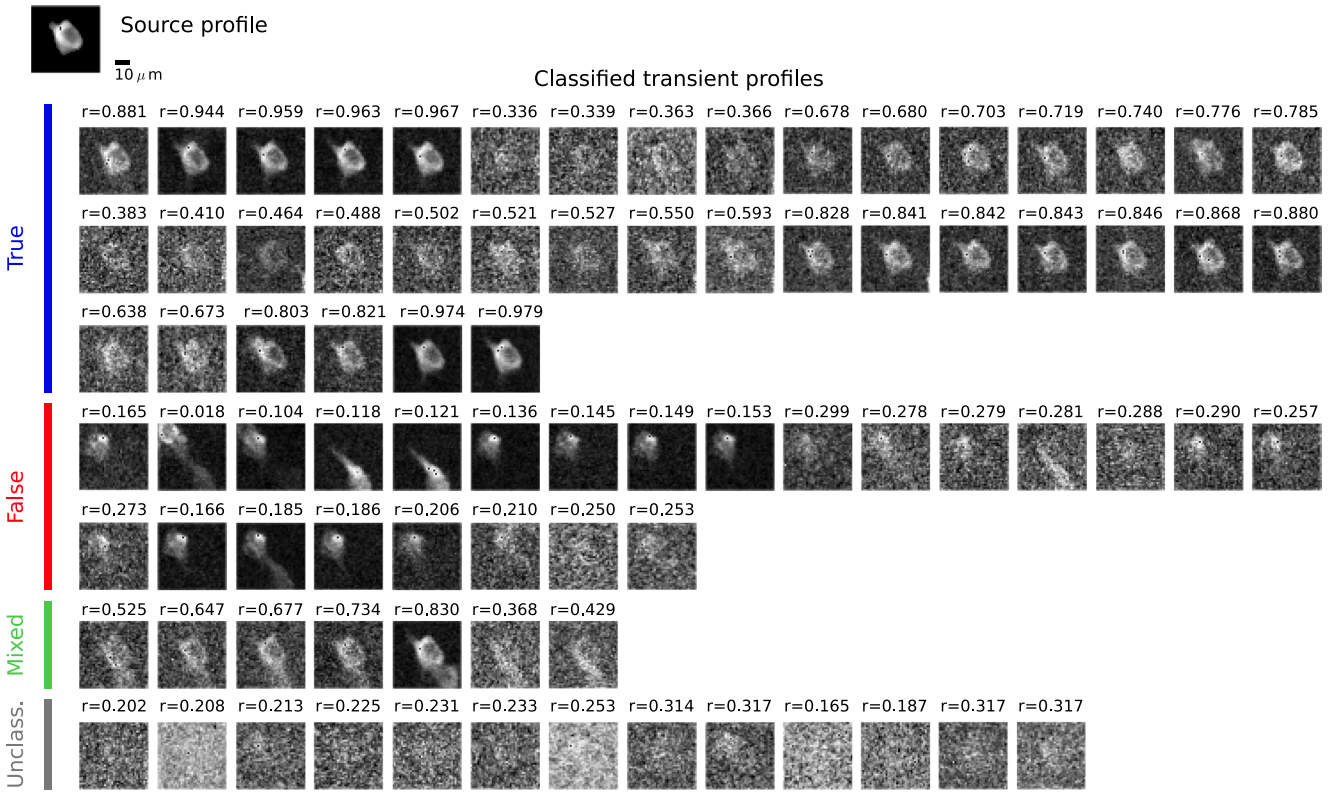
**Extended data** is available for this paper at https://doi.org/10.1038/s41592-022-01422-5.

**Supplementary information** The online version contains supplementary material available at https://doi.org/10.1038/s41592-022-01422-5.

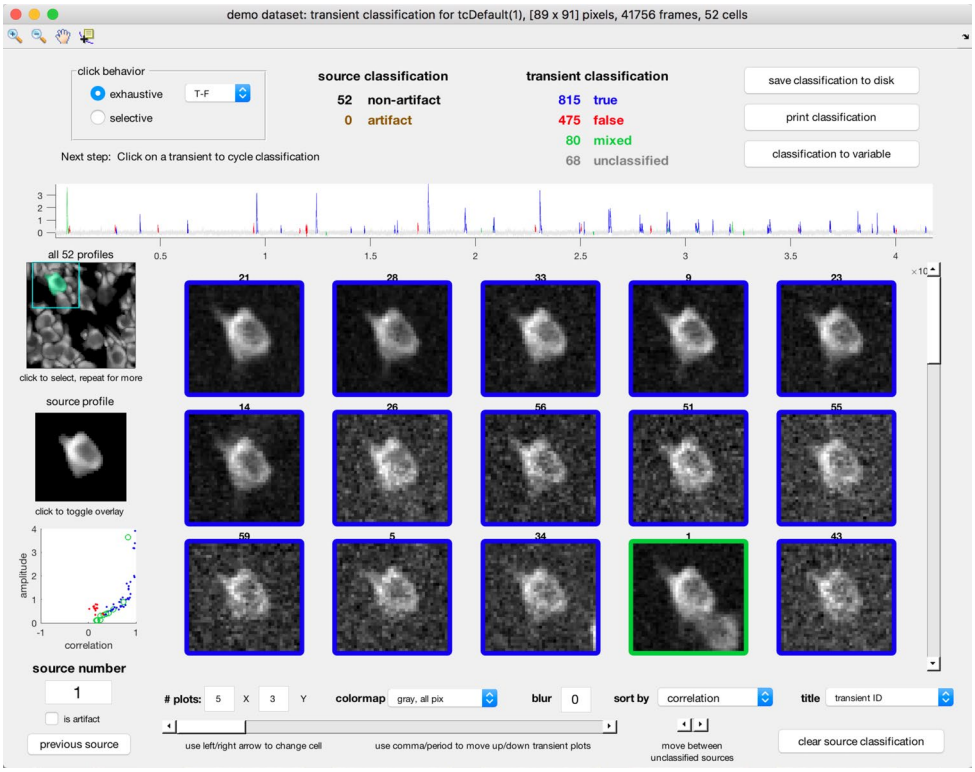
**Correspondence and requests for materials** should be addressed to David W. Tank or Adam S. Charles.

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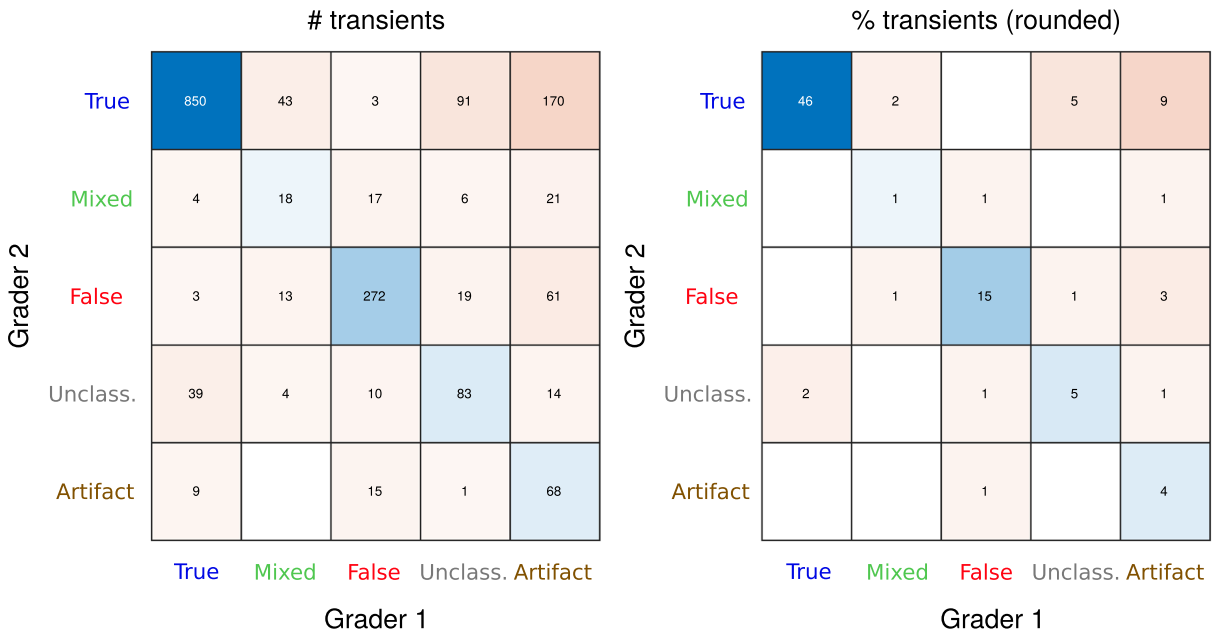
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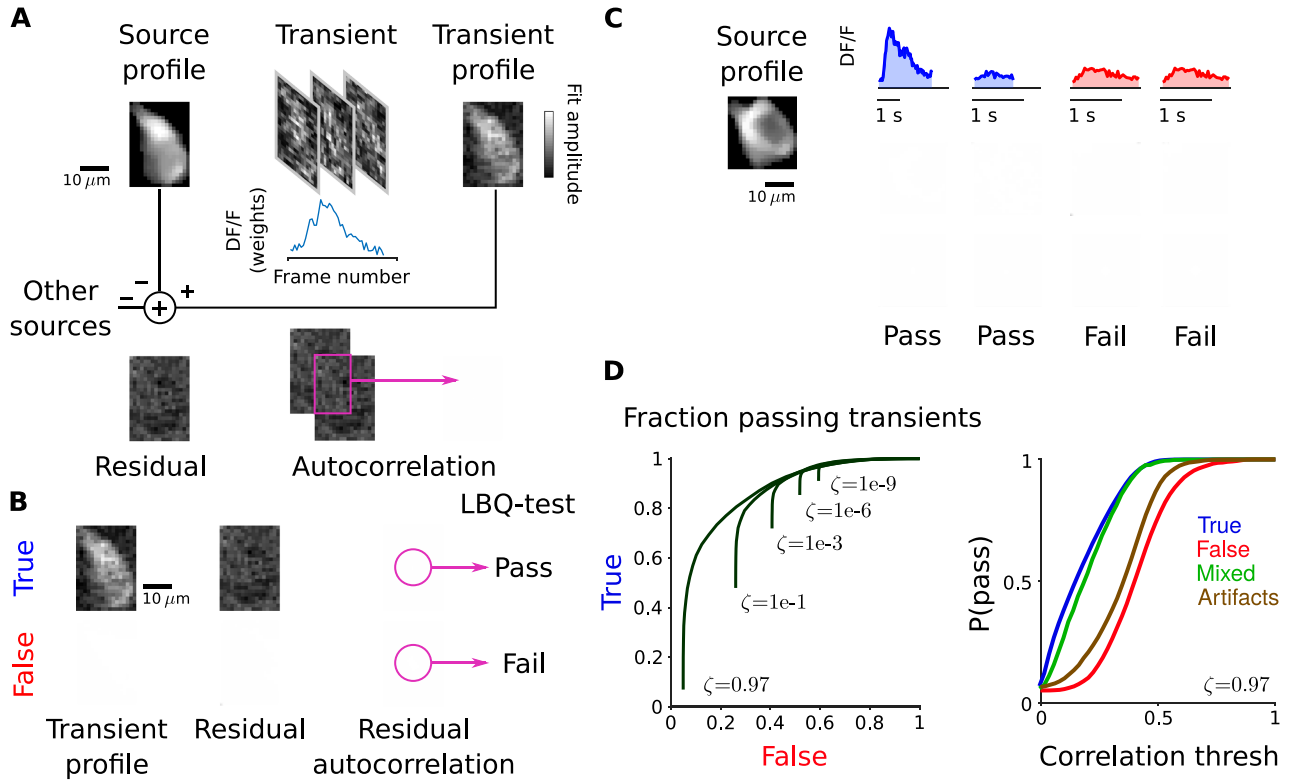

**Extended Data Fig. 1 | Example classified transient profiles for a single source from mouse CA1 found using CNMF.** Each panel is titled with the Pearson correlation between the source profile and transient profile.



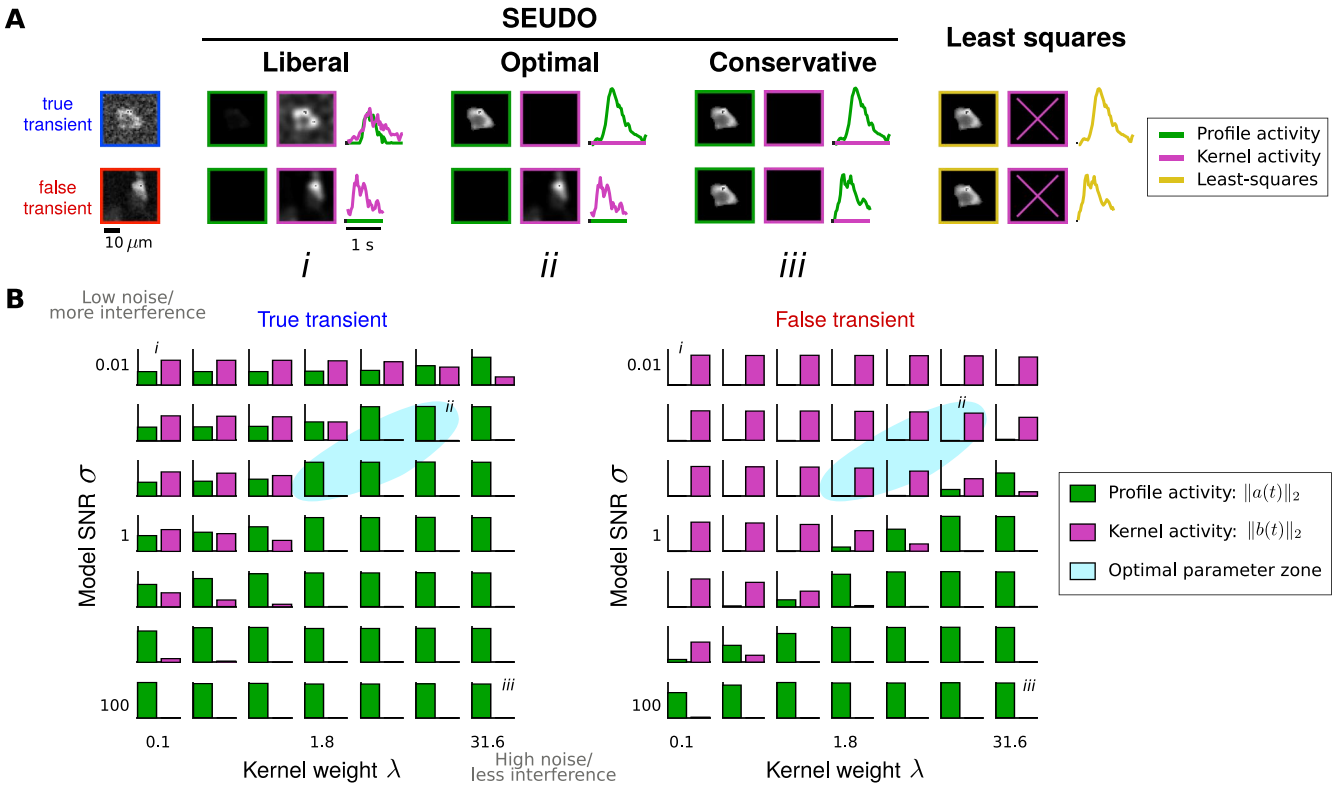

**Extended Data Fig. 2 | User interface for manual classification of transients.** For detailed description, see software documentation in the GIT repository https://github.com/adamshch/SEUDO.



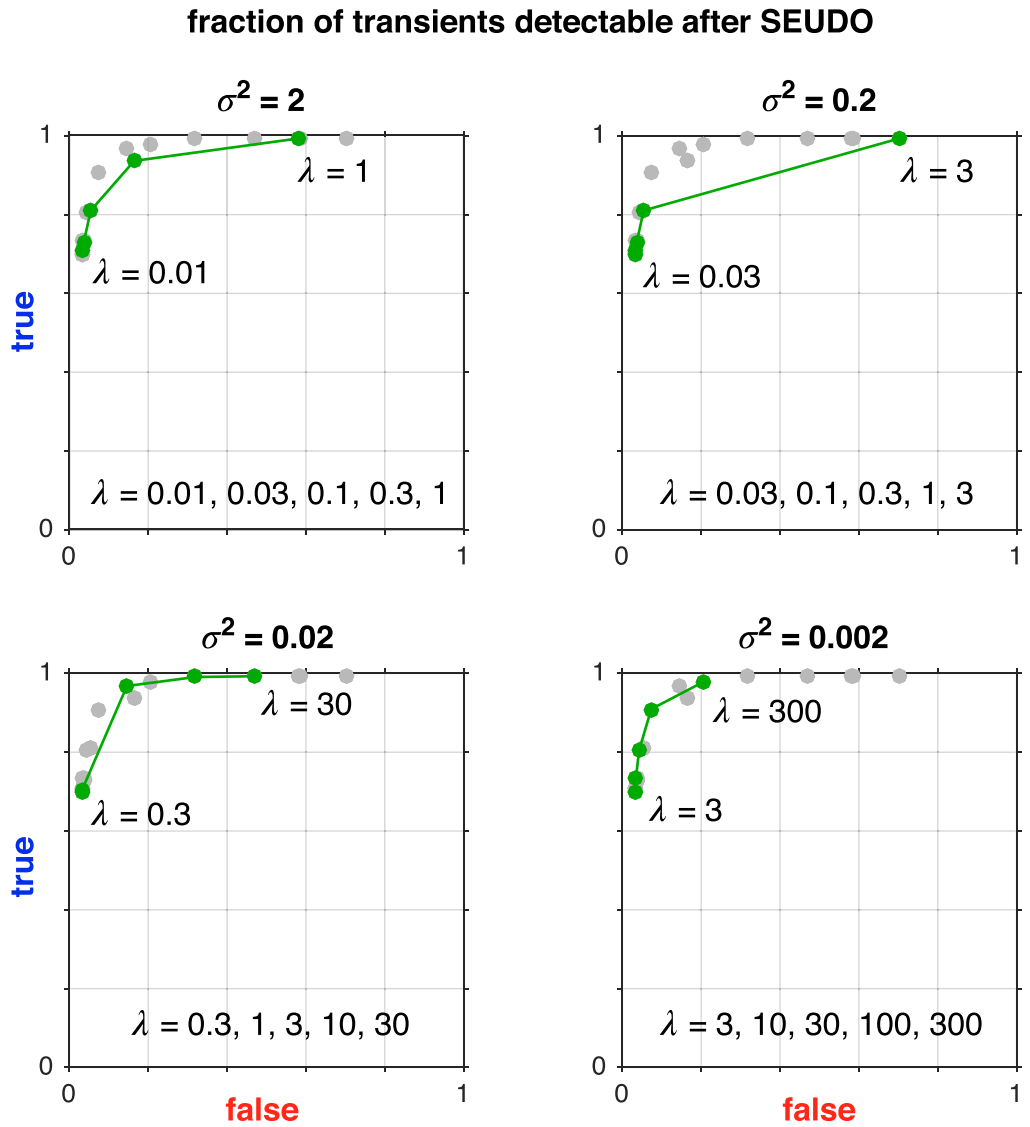

**Extended Data Fig. 3 | Comparison of manual annotation for two graders on the same 1,834 transients.** Confusion matrices showing the number of transients graded in different ways by each reviewer (left) and the total percentage of transients in each bin (right).



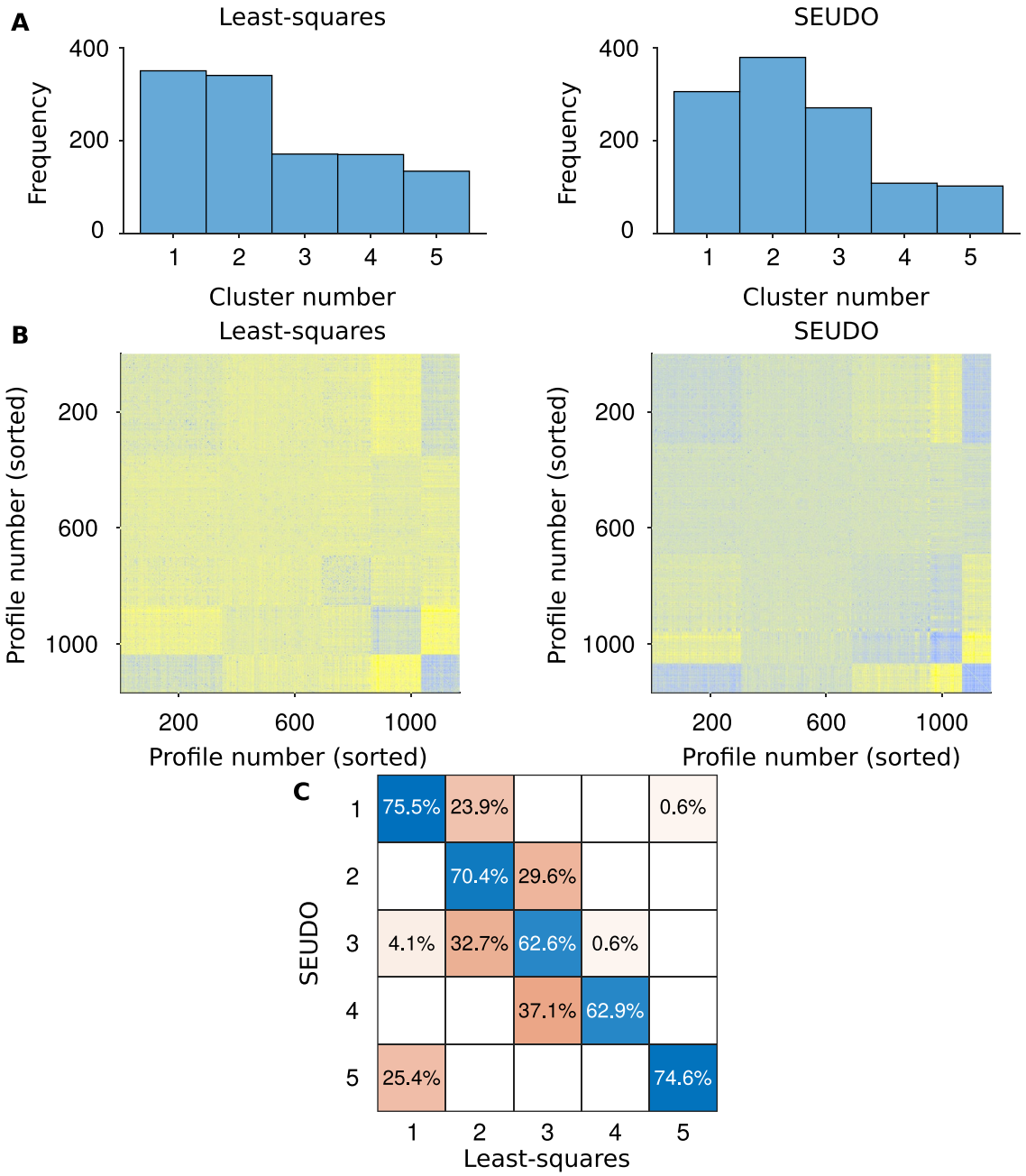

**Extended Data Fig. 4 | Algorithm to classify transients as true or false using the spatial Ljung-Box quartile test instead of the correlation metric.** A: Schematic for how the transient profile residual autocorrelation is computed. B: Two example transients, shown to illustrate difference in residual for true and false transients. C: Classification of four example transients using the LBQ test. D: Results of the LBQ test on transients classified by human expert. Left: Results of the test applied to true and false transients for various values of $\alpha$. Right: Results of the test applied to all four transient types.




**Extended Data Fig. 5 | Effect of SEUDO parameters.** A: Activity estimated by SEUDO and least squares for one true transient (top row) and one false transient (bottom row) using several types of estimation (column labels). Images and traces show estimated amplitude of the source profile (green) and sum of fitted Gaussian kernels (magenta). B: Sum of activity ascribed to the source profile (green) and Gaussian kernels (magenta) for the true transient (left) and the false transient (right). Each subplot shows results for one set of parameters. Roman numerals indicate parameter regimes shown in A.




**Extended Data Fig. 6 | Performance of SEUDO as $\sigma^2$ was varied over three orders of magnitude, for the same sources and quantified in the same way as in Fig. 5c.** Each subplot shows performance for the value of $\sigma^2$ indicated in the title and the shown values of $\lambda$ (green points). Also plotted for comparison are the collection of points taken from all subplots (gray points).




**Extended Data Fig. 7 | Removing false transients can impact global summaries of activity.** Here, time courses were sorted into 5 clusters using K-means (best clustering over 50 random seeds). A-B: SEUDO time courses were more highly consolidated (that is, fewer clusters contained more of the neurons) as compared to the time-traces with contamination. C: A confusion matrix depicts that a small number of cluster relabelings accounted for much of the changes in SEUDO time courses.

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For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

| n/a | Confirmed | |
|---|---|---|
| ☐ | ☒ | The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement |
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| ☒ | ☐ | A description of all covariates tested |
| ☒ | ☐ | A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons |
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| ☒ | ☐ | For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ | For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☐ | ☒ | Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated |

*Our web collection on statistics for biologists contains articles on many of the points above.*

## Software and code

Policy information about availability of computer code

| | |
|---|---|
| Data collection | Calcium imaging data was collected with the proprietary software package ScanImage, version 5 (Invidrio). |
| Data analysis | Motion correction was performed with either the open source software package Caiman (2016 version) or using custom code using a rigid correlation-based method. Identification of cell profiles was performed using the open source software package CNMF (2016 version, now included in Caiman package) and the open source Suite2p package for Matlab maintained by Janelia Research Institute (version from Apr 2, 2019) . Time trace estimation was performed with custom software (available at https://github.com/adamshch/SEUDO), some of which used the TFOCS package (version 1.4.1). |

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Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

The data that support the findings of this study are available on the Open Science Framework at https://osf.io/zt54q

# Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

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# Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

| | |
|---|---|
| Sample size | No sample size calculation was performed. Specific two-photon calcium imaging movies were selected to be representative of different brain areas, labeling densities, and SNR levels. Each dataset contained hundreds or thousands of transients, enough to estimate the approximate rate of contamination in equivalent datasets. |
| Data exclusions | No data were excluded. |
| Replication | Our findings were upheld in all datasets across a wide range of conditions, and expert human graders produced substantially similar scoring. |
| Randomization | Randomization was not relevant. Since all analyses were performed on the same set of sources, the results could not be explained by treatment groups having different compositions. |
| Blinding | When manually classifying transients, the human classifiers were blinded to the results of automatic analysis and to each other's classification. For imaging experiments, investigators were not blinded to group allocation since it would not be expected to affect the acquired images. |

# Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

## Materials & experimental systems

| n/a | Involved in the study |
|---|---|
| ☒ | ☐ Antibodies |
| ☒ | ☐ Eukaryotic cell lines |
| ☒ | ☐ Palaeontology and archaeology |
| ☐ | ☒ Animals and other organisms |
| ☒ | ☐ Human research participants |
| ☒ | ☐ Clinical data |
| ☒ | ☐ Dual use research of concern |

## Methods

| n/a | Involved in the study |
|---|---|
| ☒ | ☐ ChIP-seq |
| ☒ | ☐ Flow cytometry |
| ☒ | ☐ MRI-based neuroimaging |

## Animals and other organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research

| | |
|---|---|
| Laboratory animals | Data were collected from male mice aged 7 to 15 weeks at the time of surgery. Three strains were used: C57BL/6J-Tg (Thy1-GCaMP3) GP2.11Dkim/J, Ai93-D; CaMKIIa-tTA Igs7tm93.1(tetO-GCaMP6f)Hze Tg(Camk2a-tTA)1Mmay/J crossed with Emx1-Cre (B6.129S2-Emx1tm1(cre)Krj/J, and a cross of Emx1-Cre B6.129S2-Emx1tm1(cre)Krj/J, CaMK2-tTA B6.Cg-Tg(Camk2a-tTA)1Mmay/DboJ, and TITL-GCaMP6f (Ai93; B6.Cg-Igs7tm93.1(tetOGCaMP6f)Hze/J. Mice were maintained on a reverse light cycle, and imaging was performed during the dark phase. Ambient temperatures were 69-72°F with 30-70% humidity. |
| Wild animals | The study did not involve wild animals. |
| Field-collected samples | No samples were collected in the field. |
| Ethics oversight | All protocols were approved by the Princeton University Institutional Animal Care and Use Committee |

Note that full information on the approval of the study protocol must also be provided in the manuscript.